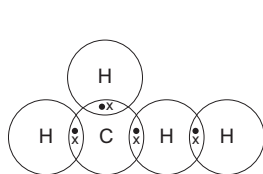
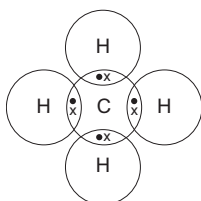
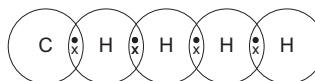
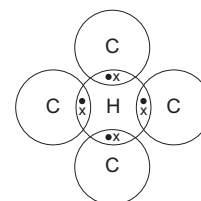


3. (a) Methane gas, CH_4 , contains the elements carbon and hydrogen.

Element	Electronic structure
carbon	2,4
hydrogen	1

- (i) I. Put a **tick** (✓) in the box below the diagram which shows how the atoms are bonded in one molecule of methane. [1]


☐

☐

☐

☐

- II. **Circle** the name given to the type of bonding found in methane. [1]

covalent

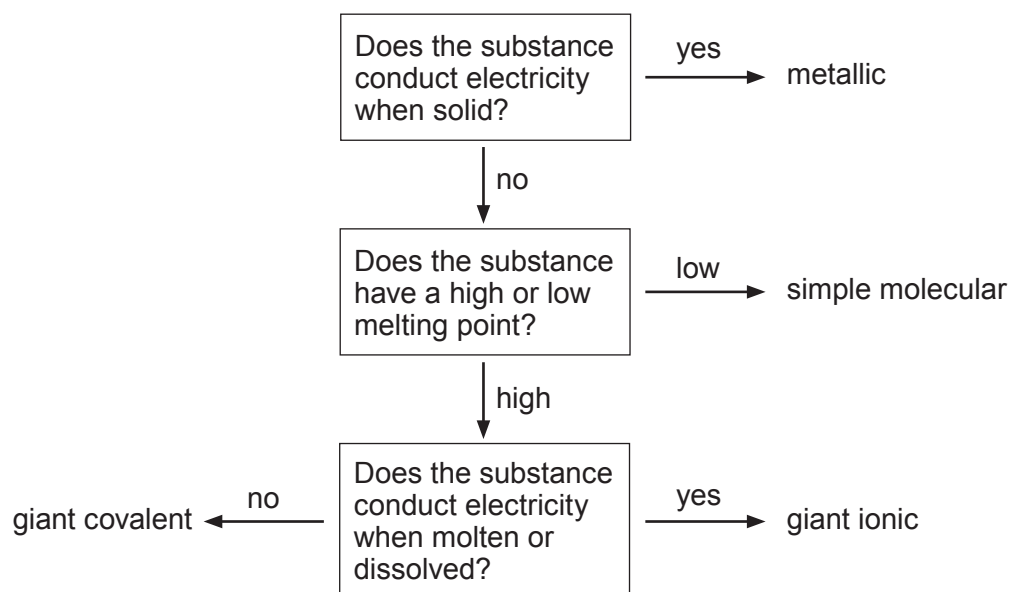
ionic

giant covalent

metallic



- (ii) The flow chart below can be used to identify the type of structure found in different substances.



- I. Use the flow chart to complete the table.

[1]

Substance	Melting point (°C)	Electrical conductivity	Type of structure
A	2072	conducts only when molten	
B	-182	does not conduct electricity	
C	1610	does not conduct electricity	

- II. Give the **letter** of the substance, **A**, **B** or **C**, that is most likely to be methane. [1]

Substance



- (b) Butene, C_4H_8 , also contains the elements carbon and hydrogen. Calculate the percentage by mass of hydrogen in butene. [2]

$$A_r(H) = 1 \quad A_r(C) = 12$$

Percentage = %

Examiner
only

6

